Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Magnetic moment of Gd^{3+} ion $(\mathrm{Z}=64)$ is

(A) $3.62\mathrm{BM}$

(B) $9.72\mathrm{BM}$

(C) $7.9\mathrm{BM}$

(D) $10.60\mathrm{BM}$

Q.2

Which of the following forms stable $+4$ oxidation state?

(A) $\mathrm{La}(\mathrm{Z}=57)$

(B) $\mathrm{Eu}(Z=63)$

(C) $\mathrm{Ce}(\mathrm{Z}=58)$

(D) $Gd(Z=64)$

Q.3

The number of unpaired electrons in Gadolinium $[Z=64]$ is

(A) 3

(B) 8

(C) 6

(D) 2

Q.4

Which one of the following elements shows maximum number of different oxidation states in its compounds?

(A) Eu

(B) La

(C) Gd

(D) Am

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Q.5

The radius of La^{3+} (Atomic number of $La = 57$) is $1.06\mathring{A}$. Which one of the following given values will be closest to the radius of Lu^{3+} (Atomic number of $Lu = 71$)?

(A) $1.40\mathring{A}$

(B) $1.06\mathring{A}$

(C) $0.85\mathring{A}$

(D) $1.60\mathring{A}$

Q.6

The electronic configuration of gadolinium (At. No. 64) is

(A) $[Xe]4f^8 5d^1 6s^2$

(B) $[Xe]4f^7 5d^1 6s^2$

(C) $[Xe]4f^3 5d^5 6s^2$

(D) $[Xe]4f^6 5d^2 6s^2$

Q.7

Which of the following characteristics of lanthanides is correct ?

(A) Reducing property decreases from La to Lu

(B) Reducing property increases from La to Lu

(C) Reducing property remains constant from La to Lu

(D) Reducing property initially decreases followed by increase

Q.8

The most of toner-transition elements, show the valency of

(A) $+2$

(B) $+3$

(C) $+4$

(D) $+5$

Q.9

Which of the following pairs is expected to form colourless compound?

(A) ${}_{57}La^{3+}$ ${}_{65}Tb^{3+}$

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(B) ${}_{57}La^{3+}$ ${}_{71}Lu^{3+}$

(C) ${}_{60}\mathrm{Nd}^{3+}$ ${}_{61}\mathrm{Pm}^{3+}$

(D) ${}_{63}\mathrm{Eu}^{3+}$ ${}_{65}\ \mathrm{Tb}^{3+}$

Q.10

Arrange $\mathrm{Ce}^{+3}, \mathrm{La}^{+3}, \mathrm{Pm}^{+3}$ and Yb^{+3} in increasing order of their ionic radii

(A) $\mathrm{Yb}^{+3} < \mathrm{Pm}^{+3} < \mathrm{Ce}^{+3} < \mathrm{La}^{+3}$

(B) $\mathrm{Ce}^{+3} < \mathrm{Yb}^{+3} < \mathrm{Pm}^{+3} < \mathrm{La}^{+3}$

(C) $\mathrm{Yb}^{+3} < \mathrm{Pm}^{+3} < \mathrm{La}^{+3} < \mathrm{Ce}^{+3}$

(D) $\mathrm{Pm}^{+3} < \mathrm{La}^{+3} < \mathrm{Ce}^{+3} < \mathrm{Yb}^{+3}$

Q.11

Which of the following trivalent ion has the largest atomic radii in the lanthanide series

(A) La

(B) Ce

(C) Pm

(D) Lu

Q.12

Lanthanum is gouped with f block elements because

(A) It has partially filled forbitals

(B) It is just before Ce in the periodic table

(C) It has both partially filled f and d-orbitals

(D) The properties of Lanthanum are very similar to the elements of $4f$ block

Q.13

Which element among the lanthandes has the smallest atomic radius?

(A) Cetium

(B) Lutetium

(C) Europium

(D) Gadolinium

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Q.14

Which of the following can act as oxidising agent?

(A) Ce^{4+}

(B) Sm^{2+}

(C) Gd^{3+}

(D) Lu^{3+}

Q.15

Which of the following species has maximum tendency to form complexes?

(A) La^{3+}

(B) Lu^{3+}

(C) Git $^{3+}$

(D) Ce^{3+}

Q.16

Transuranic elements begin with

(A) Np

(B) Ca

(C) Pu

(D) U

Q.17

In^{+3} (trivalent lanthanides ions) have E.C.

(A) $[Xe]4f^{1}to(Xe)4f^{14}$

(B) $[Xe]4\ d^{1}4f^{I}$ to $[Xe]4\ d^{1}4f^{4}$

(C) $[Xe]4\ d^{2}4f^{0}$ to $[Xe]4\ d^{1}4^{4}t^{4}$

(D) $[Xe]4f^{\circ}$ to $[Xe]4f^{4}$

Q.18

The colour of lanthanides and actinides is due to:

(A) $s-f$ transitions

(B) $d-f$ transitions

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(C) $\mathrm{p}-\mathbf{f}$ transitions

(D) $\mathrm{f}-\mathrm{f}$ transition

Q.19

Which group of metals is called uranides:

(A) $\mathrm{Fe}, \mathrm{Co}, \mathrm{Ni}$

(B) Ce, Pr, Nd

(C) $\mathrm{Ac}, \mathrm{Th}, \mathrm{Pa}$

(D) $\mathrm{Pu}, \mathrm{Am}, \mathrm{Cm}$

Q.20

Across lanthanide series the basicity of lanthanide hydroxides

(A) Decreases

(B) Increases

(C) First decreases and then increases

(D) First increases and then decreases

Answers & Solutions

Q.1 Answer:

$7.9\mathrm{BM}$

Solution:

Gd $(64) = [\mathrm{Xe}]4f^7 5d^1 6s^2$

$\mathrm{Gd}^{3+} = [\mathrm{Xe}]4f^7 5d^0 6s^0$

i.e. no. of unpaired electrons $= 7$

$$\mu = \sqrt{n(n+2)} = \sqrt{7(7+2)} = \sqrt{63} = 7.93\mathrm{BM}$$

Q.2 Answer:

$\mathrm{Ce}(\mathrm{Z}=58)$

Solution:

Only $\mathrm{Ce}(\mathrm{Z}=58)$ **shows stable** $(+4)$ **oxidation state.**

Q.3 Answer:

8

Solution:

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$\mathrm{Gd}(64) = [\mathrm{Xe}]4f^7 5d^1 6s^2$

$\therefore$ **No. of unpaired electrons** $= 8$

Q.4 Answer:

Am

Solution:

We know that lanthanides La, Gd shows $+3$, oxidation state, while Eu shows oxidation state of $+2$ and $+3$. Am shows $+3$, $+4$, $+5$ and $+6$ oxidation states. Therefore Americium (Am) has maximum number of oxidation states.

Q.5 Answer:

$0.85\mathring{A}$

Solution:

Ionic radii $\propto \frac{1}{z}$

Thus, $\frac{z_2}{z_1} \Rightarrow \frac{1.06}{(\text{ Ionic radii of } \mathrm{Lu}^{3+})} = \frac{71}{57}$

$\Rightarrow$ **Ionic radii of** $\mathrm{Lu}^{3+} = 0.85\mathring{A}$

Q.6 Answer:

$[Xe]4f^7 5d^1 6s^2$

Solution:

Electronic configuration of gadolinium is $[\mathrm{Xe}]4f^7 5d^1 6s^2$

Q.7 Answer:

Reducing property decreases from La to Lu

Solution:

E° **value increases (becomes less negative) from La to Lu. Thus, the tendency of the reaction** $M \rightarrow \mathrm{M}^{3+} +$ **3c decreases.**

Q.8 Answer:

$+3$

Solution:

Most of inner-transition elements show the valency of $+3$**.**

Q.9 Answer:

$_{57}La^{3+} \quad _{71}Lu^{3+}$

Solution:

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The configurations of $_{57}La^{3+}$ is (4f)" while that of $_{I}Lx^{3+}$ is $(4f)^{14}$. They are expected to be colomicis.

Q.10 Answer:

$Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

Solution:

Dre to Lanthanoid contraction order will be $Yb^{+3} < Pm^{3+} < Ce^{+3} < La^{3+}$

Q.11 Answer:

La

Solution:

Lanthanum is the first element of lanthanide series so size decreases with increase in atomic number so La has the largest atomic radi.

Q.12 Answer:

It has partially filled forbitals

Solution:

It has partially filled f-orbitals. because, forbitals of electron shell are filled with " $n-2$ ". and deu to electronic configuration.

Q.13 Answer:

Lutetium

Solution:

Lutetium is the last element of lanthanides. Due to lanthanide contraction it has the smallest size.

Q.14 Answer:

Ce^{4+}

Solution:

Ce ^{4} has tendency to go over to Ce^{3+} and hence acts as oxidiring agent.

Q.15 Answer:

Lu^{3+}

Solution:

Lu^{3+} being smallest (due to lanthanide contraction) has maximum tendency to form complexes.

Q.16 Answer:

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Np

Solution:

Transuranic elements are members of the actinide series beyond uranium, beginning with neptunium (atomic number 93). The last in the series is element 103 (lawrencium). All are artificially produced in nuclear reactors, accelerators, or explosions of nuclear weapons, and all have several isotopes that emit alpha rays.

Q.17 Answer:

$[\mathrm{Xe}]4f^{1}\mathrm{to}(\mathrm{Xe})4f^{14}$

Solution:

Lanthanides have configuration of $[\mathrm{Xe}]4\ \mathrm{F}^{1-14}5\ \mathrm{d}^{1}6\ \mathrm{s}^{2}$
So their trivalent ion have configuration of $[\mathrm{Xe}]4\mathrm{f}^{1}$ **to** $[\mathrm{Xe}]4\ \mathrm{F}^{14}$

Q.18 Answer:

$\mathrm{f}-\mathrm{f}$ transition

Solution:

No solution available

Q.19 Answer:

$\mathrm{Pu}, \mathrm{Am}, \mathrm{Cm}$

Solution:

Pu, Am and Cm **are prepared from uranium, hence called urenides.**

Q.20 Answer:

Decreases

Solution:

Across the lanthanide series the size of the cations decreases thereby decreasing ionic character and increasing covalent character of hydroxides of lanthanides. Thus, the basicity of hydroxides decreases.

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